

Chemical Equilibrium and K_c

K_c expressions, Q versus K and ICE tables

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$$a A + b B \rightleftharpoons c C + d D: K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b} \text{ (each concentration divided by } c^\circ = 1 \text{ M)}$$

Leave out pure solids and liquids · Reverse: $1/K$ · Multiply coefficients by n : K^n

$Q < K$: forward · $Q > K$: reverse · $Q = K$: at equilibrium

Write concentrations with units (M) and give K_c to three significant figures.

A. Understanding

1. Explain what is meant by a dynamic equilibrium. Are the concentrations of reactants and products equal at equilibrium?

B. Writing K_c

2. Write the K_c expression for: (a) $N_2(g) + 3 H_2(g) \rightleftharpoons 2 NH_3(g)$ (b) $2 SO_2(g) + O_2(g) \rightleftharpoons 2 SO_3(g)$ (c) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ (d) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

C. Calculating K_c

3. An equilibrium mixture of $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ contains 0.0400 M SO_2 , 0.0200 M O_2 and 0.160 M SO_3 . Calculate K_c .

4. Using your answer to question 3, find K_c for (a) $2\text{SO}_3 \rightleftharpoons 2\text{SO}_2 + \text{O}_2$ and (b) $\text{SO}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{SO}_3$.

D. Q versus K

5. For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_c = 0.0200$. A mixture contains 0.100 M N_2O_4 and 0.0200 M NO_2 . Calculate Q and predict the direction of change.

E. ICE tables

6. 0.200 M H_2 and 0.200 M I_2 are mixed: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $K_c = 49.0$. Find the equilibrium concentrations of all three gases.

7. 0.500 M PCl_5 is placed in a container: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. At equilibrium, $[\text{Cl}_2] = 0.100 \text{ M}$. Calculate K_c .

F. Challenge

8. One reaction has $K = 1 \times 10^{10}$ and another $K = 1 \times 10^{-10}$. Describe the equilibrium mixture in each case.

9. Using K_c from question 7, find the equilibrium concentrations when 0.300 M PCl_5 is placed in a container. (You will need to solve a quadratic equation.)